Chapter 10 Definite Integrals: Examples Worksheet

Solutions (teacher reference)

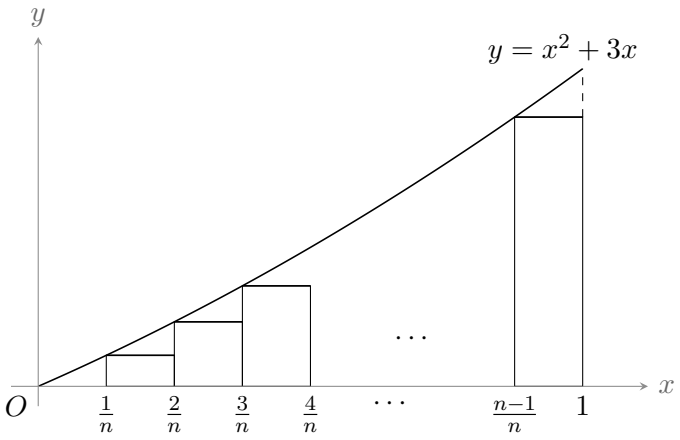
JC and A-level examination questions (H2 Mathematics 9758): three on the limit of a sum for Section 10.1, then one for each later section (Promotional Examination where available, otherwise Preliminary Examination). Questions are reproduced as set, with the original marks.

Q	Section	Source
1	10.1 The definite integral as a limit of a sum	DHS/2019/P1/Q10†
2	10.1 The definite integral as a limit of a sum	ACJC/2019/P1/Q6†
3	10.1 The definite integral as a limit of a sum	9758/2015/P1/Q3†
4	10.2 Area between a curve and the x-axis	NJC 2024 Promo Q3
5	10.3 Area between a curve and the y-axis	SAJC 2025 Prelim P1 Q7
6	10.4 Area between two curves	YIJC 2025 Prelim P1 Q12
7	10.5 Volume of revolution	NJC 2024 Promo Q2
8	10.6 Parametric curves: area	VJC 2025 Promo Q11(c), adapted

†From the Section 1 exercises in the teacher's own Definite Integrals notes.

Question 1 10.1

[DHS/2019/P1/Q10]



The diagram above shows part of the graph of $y = x^2 + 3x$, with rectangles approximating the area under the curve from $x = 0$ to $x = 1$. The area under the curve may be approximated by the total area, A, of $(n-1)$ rectangles each of width $\frac{1}{n}$. Given that

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1),$$

show that $A = \dfrac{(n-1)(11n-1)}{6n^2}$.

Explain briefly how the value of $\displaystyle\int_0^1 x^2 + 3x \ dx$ can be deduced from this expression, and hence find this value exactly without integration. **[6]**

Solution

With $\mathrm{f}(x) = x^2 + 3x$:

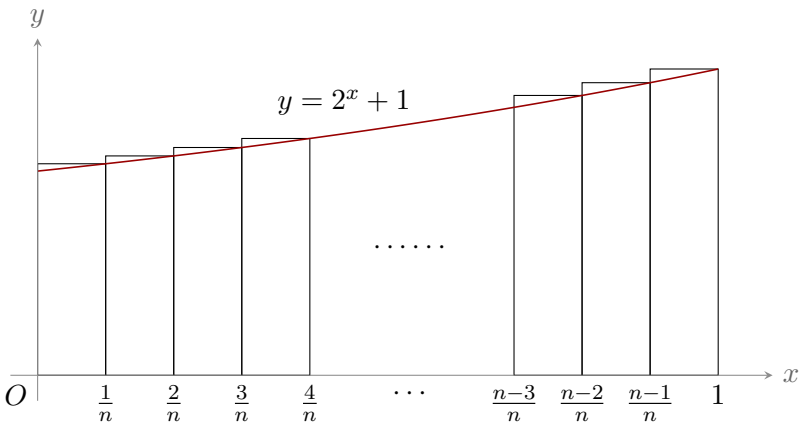
$$\begin{aligned}
A &= \frac{1}{n}\left[\left(\frac{1}{n}\right)^2 + 3\left(\frac{1}{n}\right)\right] + \frac{1}{n}\left[\left(\frac{2}{n}\right)^2 + 3\left(\frac{2}{n}\right)\right] + \frac{1}{n}\left[\left(\frac{3}{n}\right)^2 + 3\left(\frac{3}{n}\right)\right] \\
&\quad + \cdots + \frac{1}{n}\left[\left(\frac{n-1}{n}\right)^2 + 3\left(\frac{n-1}{n}\right)\right] \\
&= \sum_{r=1}^{n-1} \frac{1}{n}\left[\left(\frac{r}{n}\right)^2 + 3\left(\frac{r}{n}\right)\right] \\
&= \sum_{r=1}^{n-1} \left(\frac{1}{n^3}r^2 + \frac{3}{n^2}r\right) \\
&= \frac{1}{n^3}\sum_{r=1}^{n-1} r^2 + \frac{3}{n^2}\sum_{r=1}^{n-1} r \\
&= \frac{1}{n^3}\left[\frac{(n-1)(n)(2n-1)}{6}\right] + \frac{3}{n^2}\left[\frac{n(n-1)}{2}\right] \\
&= \frac{(n-1)(2n-1)}{6n^2} + \frac{3(n-1)}{2n} \\
&= \frac{(n-1)(2n-1) + 9n(n-1)}{6n^2} \\
&= \frac{(n-1)(2n-1+9n)}{6n^2} \\
&= \frac{(n-1)(11n-1)}{6n^2} \text{ (shown)}
\end{aligned}$$

As $n \to \infty$, the total area of the rectangles approaches the actual area under the curve, which is $\int_0^1 \left(x^2 + 3x\right) dx$.

$$\begin{aligned}
\int_0^1 \left(x^2 + 3x\right) dx &= \lim_{n\to\infty} \frac{(n-1)(11n-1)}{6n^2} \\
&= \lim_{n\to\infty} \frac{11n^2 - 12n + 1}{6n^2} \\
&= \lim_{n\to\infty} \frac{11 - \frac{12}{n} + \frac{1}{n^2}}{6} \\
&= \frac{11}{6}
\end{aligned}$$

Question 2 10.1 [ACJC/2019/P1/Q6]

The diagram below shows the graph of $y = 2^x + 1$ for $0 \le x \le 1$. Rectangles, each of width $\frac{1}{n}$, are also drawn on the graph as shown.



Show that the total area of all n rectangles, S_n, is given by

$$S_n = \frac{2^{1/n}}{n\left(2^{1/n} - 1\right)} + 1.$$

[3]

Find the exact value of $\lim_{n\to\infty} S_n$. **[2]**

Solution

Show that:

$$\begin{aligned}
S_n &= \frac{1}{n}\left(2^{\frac{1}{n}} + 1\right) + \frac{1}{n}\left(2^{\frac{2}{n}} + 1\right) + \frac{1}{n}\left(2^{\frac{3}{n}} + 1\right) \\
&\quad + \cdots + \frac{1}{n}\left(2^{\frac{n}{n}} + 1\right) \\
&= \sum_{r=1}^{n} \frac{1}{n}\left(2^{\frac{r}{n}} + 1\right) \\
&= \frac{1}{n}\sum_{r=1}^{n} 2^{\frac{r}{n}} + \frac{1}{n}\sum_{r=1}^{n} 1 \\
&= \frac{1}{n}\sum_{r=1}^{n} \left(2^{\frac{1}{n}}\right)^r + \frac{1}{n}(n) \\
&= \frac{1}{n}\left[\frac{2^{\frac{1}{n}}\left[\left(2^{\frac{1}{n}}\right)^n - 1\right]}{2^{\frac{1}{n}} - 1}\right] + 1 \\
&= \frac{1}{n}\left[\frac{2^{\frac{1}{n}}(2-1)}{2^{\frac{1}{n}} - 1}\right] + 1 \\
&= \frac{2^{\frac{1}{n}}}{n\left(2^{\frac{1}{n}} - 1\right)} + 1 \text{ (shown)}
\end{aligned}$$

Sum of a GP: $\sum_{r=1}^{n} a^r = \frac{a\left(a^n - 1\right)}{a - 1}$

Alternatively: $2^{\frac{1}{n}}$, $2^{\frac{2}{n}}$, $2^{\frac{3}{n}}$, … is a GP; each term is $2^{\frac{1}{n}}$ times the one before.

$$\begin{aligned}
S_n &= \tfrac{1}{n}\left(2^{\frac{1}{n}} + 2^{\frac{2}{n}} + 2^{\frac{3}{n}} + \cdots + 2^{\frac{n}{n}}\right) \\
&\quad + \tfrac{1}{n}\left(1 + 1 + 1 + \cdots + 1\right) \\
&= \tfrac{1}{n}\left(\frac{2^{\frac{1}{n}}\left[\left(2^{\frac{1}{n}}\right)^n - 1\right]}{2^{\frac{1}{n}} - 1}\right) + \tfrac{1}{n}(n) \\
&= \vdots \text{ (as on the left)}
\end{aligned}$$

Limit:

$$\begin{aligned}
\lim_{n\to\infty} S_n &= \int_0^1 \left(2^x + 1\right) dx \\
&= \left[\frac{2^x}{\ln 2} + x\right]_0^1 \\
&= \frac{2}{\ln 2} + 1 - \frac{1}{\ln 2} \\
&= \frac{1}{\ln 2} + 1
\end{aligned}$$

Question 3 10.1 [9758/2015/P1/Q3]

(i) Given that f is a continuous function, explain, with the aid of a sketch, why the value of

$$\lim_{n\to\infty}\frac{1}{n}\left\{\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right\}$$

is $\displaystyle\int_0^1 \mathrm{f}(x)\,dx$. **[2]**

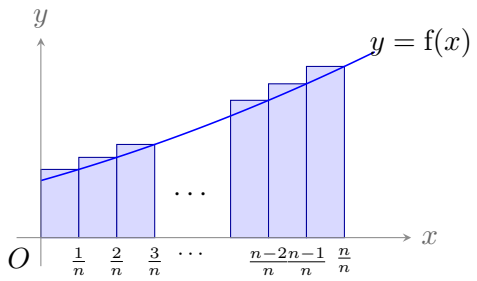
(ii) Hence evaluate $\displaystyle\lim_{n\to\infty}\frac{1}{n}\left(\frac{\sqrt[3]{1}+\sqrt[3]{2}+\cdots+\sqrt[3]{n}}{\sqrt[3]{n}}\right)$. **[3]**

Solution

(i)

Total area of rectangles

$$=\frac{1}{n}\mathrm{f}\left(\frac{1}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{2}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\frac{1}{n}\mathrm{f}\left(\frac{n}{n}\right)$$
$$=\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]$$



As $n\to\infty$, the total area of the rectangles approaches the actual area under the graph, which is $\displaystyle\int_0^1 \mathrm{f}(x)\,dx$.

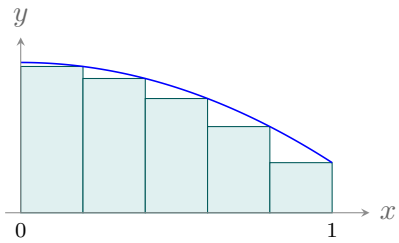
$$\therefore\ \lim_{n\to\infty}\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]=\int_0^1 \mathrm{f}(x)\,dx \text{ (shown)}$$

(ii)

$$\lim_{n\to\infty}\frac{1}{n}\left(\sqrt[3]{\frac{1}{n}}+\sqrt[3]{\frac{2}{n}}+\sqrt[3]{\frac{3}{n}}+\cdots+\sqrt[3]{\frac{n}{n}}\right)=\int_0^1 \sqrt[3]{x}\,dx$$
$$=\int_0^1 x^{\frac{1}{3}}\,dx$$
$$=\left[\frac{x^{\frac{4}{3}}}{\frac{4}{3}}\right]_0^1$$
$$=\frac{3}{4}$$

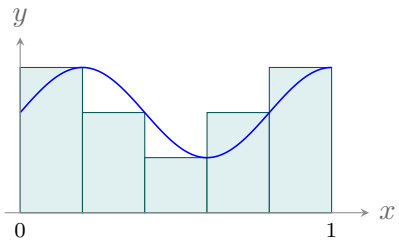
Question 3: right-hand or left-hand rectangles? (5 rectangles on $[0,1]$)

Decreasing, right-hand ✓



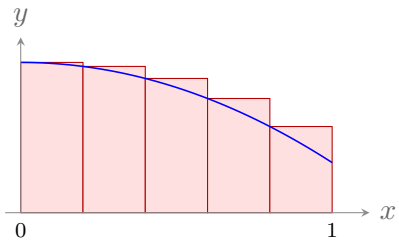
rectangles $= 1.06$, area $= 1.167$: underestimate

Up and down, right-hand ✓



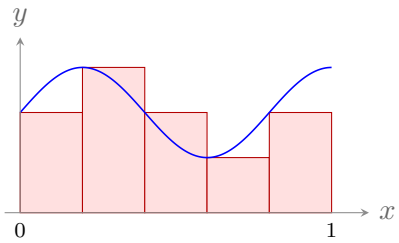
rectangles $= 1.09$, area $= 1.057$: overestimate

Decreasing, left-hand ✗



rectangles $= 1.26$, area $= 1.167$: overestimate

Up and down, left-hand ✗



rectangles $= 1.00$, area $= 1.057$: underestimate

What matters is that the rectangles use the **right-hand sum** $\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]$, as the question specifies, not whether it is an under- or overestimate. It is not correct to say this question only accepts overestimates: a right-hand sum alone does not determine under or over (top row), and a left-hand sum $\frac{1}{n}\left[\mathrm{f}\left(\frac{0}{n}\right)+\cdots+\mathrm{f}\left(\frac{n-1}{n}\right)\right]$ is not what the question describes, even when it overestimates (bottom row).

Question 4 10.2

[NJC 2024 Promo Q3]

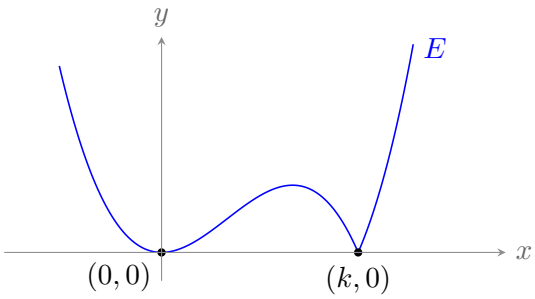
(i) Given that k is a positive constant, sketch the curve E with equation $y = \left|x^2(x-k)\right|$, labelling the coordinates of the points where the curve cuts the axes. **[2]**

(ii) Hence, find, in terms of k, the area of the region(s) bounded by E and the x-axis for $0 \leq x \leq 2k$. **[3]**

Solution

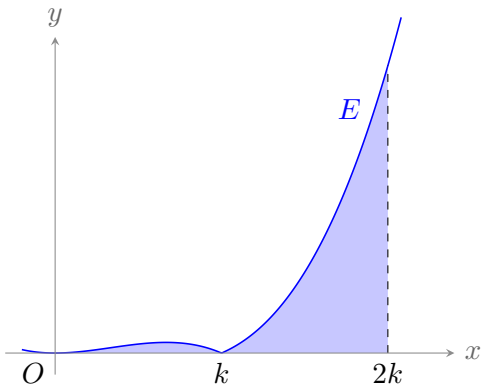
Working follows the school's solution on grail.moe: https://grail.moe/library/16719/njc-9758-2024-promo-solutions

(i)



(ii)

$$\begin{aligned}
&\int_0^{2k} \left|x^2(x-k)\right| dx \\
&= \int_0^{2k} \left|x^3 - kx^2\right| dx \\
&= -\int_0^{k} x^3 - kx^2 \, dx + \int_k^{2k} x^3 - kx^2 \, dx \\
&= -\left[\frac{x^4}{4} - k\frac{x^3}{3}\right]_0^{k} + \left[\frac{x^4}{4} - k\frac{x^3}{3}\right]_k^{2k} \\
&= \frac{3}{2}k^4
\end{aligned}$$



Question 5 10.3

[SAJC 2025 Prelim P1 Q7]

(a) (i) On a single diagram, sketch the graphs of $y = \dfrac{3x}{x-1}$ and $y = 3x$, stating the equation(s) of any asymptote(s) and the coordinates of points of intersection. **[4]**

(ii) Hence find the area bounded by the curve $y = \dfrac{3x}{x-1}$ and the lines $y = 3x$ and $y = 8$. **[2]**

(b) A curve has the equation $y = \sqrt{x^2+1}$ where x is non-negative.

(i) With the aid of a sketch of the above curve, explain why the value of

$$S = \lim_{n\to\infty} \frac{1}{n}\left[1 + \sqrt{\frac{1}{n^2}+1} + \sqrt{\frac{4}{n^2}+1} + \ldots + \sqrt{\frac{(n-1)^2}{n^2}+1}\right] \text{ is } \int_0^1 \sqrt{x^2+1}\,dx.$$

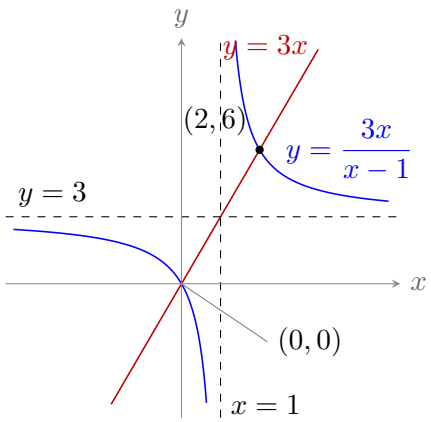
[4]

(ii) Hence find the value of S. **[1]**

Solution

Working follows the school's solution on grail.moe: https://grail.moe/library/19502/sajc-9758-2025-prelim-p1-solutions

(a)(i)



(a)(ii)

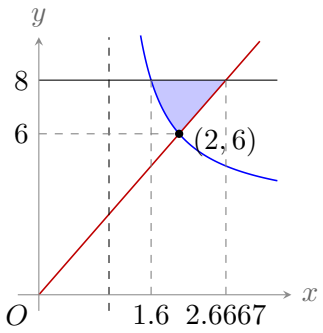
Area of region bounded by both curves and $y = 8 = \displaystyle\int_6^8 \left(\frac{y}{3} - \frac{y}{y-3}\right) dy$

Using GC,

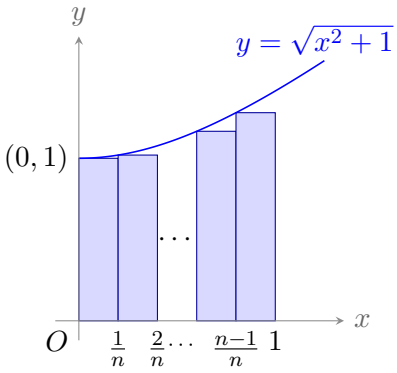
$$\text{Area} = 1.13 \text{ units}^2 \text{ (3 s.f.)}$$

Alternative method:

$$\begin{aligned}\text{Area} &= (2.6667 - 1.6)(8) - \int_{1.6}^{2} \frac{3x}{x-1}\,dx - \int_2^{2.6667} 3x\,dx \\ &= 8.5336 - 2.7325 - 4.6669 \\ &= 1.13 \text{ units}^2 \text{ (3 s.f.)}\end{aligned}$$



(b)(i) If we split the area of the region enclosed by the curve $y = \mathrm{f}(x) = \sqrt{x^2+1}$, the x-axis and the lines $x = 0$ and $x = 1$ into n rectangles, each of width $\dfrac{1}{n}$ as shown, the sum of the area of the n rectangles is given by



$$\begin{aligned} A &= \frac{1}{n}\mathrm{f}(0) + \frac{1}{n}\mathrm{f}\left(\frac{1}{n}\right) + \frac{1}{n}\mathrm{f}\left(\frac{2}{n}\right) + \cdots + \frac{1}{n}\mathrm{f}\left(\frac{n-1}{n}\right) \\ &= \frac{1}{n}\left[\mathrm{f}(0) + \mathrm{f}\left(\frac{1}{n}\right) + \mathrm{f}\left(\frac{2}{n}\right) + \cdots + \mathrm{f}\left(\frac{n-1}{n}\right)\right] \\ &= \frac{1}{n}\left[1 + \sqrt{\frac{1}{n^2}+1} + \sqrt{\frac{4}{n^2}+1} + \cdots + \sqrt{\frac{(n-1)^2}{n^2}+1}\right] \end{aligned}$$

As the number of rectangles increases, i.e. $n \to \infty$, the width of each rectangle decreases, then the sum of the area of the n rectangles approaches the area of the region enclosed by the curve $y = \sqrt{x^2+1}$, the x-axis and the lines $x = 0$ and $x = 1$, which is given by $\displaystyle\int_0^1 \sqrt{x^2+1}\,dx$.

Hence

$$\begin{aligned} S &= \lim_{n\to\infty} A \\ &= \int_0^1 \mathrm{f}(x)\,dx \\ &= \int_0^1 \sqrt{1+x^2}\,dx \end{aligned}$$

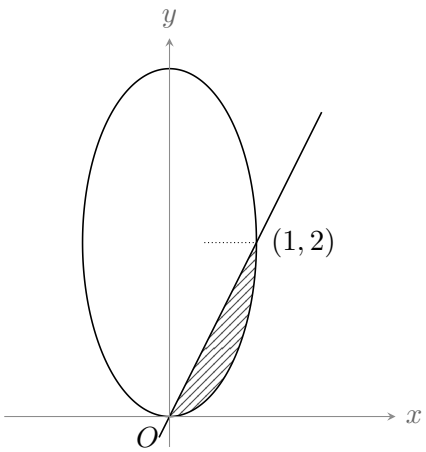
(b)(ii) Actual area under the graph $y = \sqrt{x^2+1}$ from $x = 0$ to $x = 1$. Using GC,

$$\begin{aligned} \int_0^1 \sqrt{x^2+1}\,dx &= 1.1478 \\ &= 1.15 \text{ (3 s.f.)} \end{aligned}$$

Question 6 10.4

[YIJC 2025 Prelim P1 Q12]

(a) Use the substitution $x = \sin\theta$ to show that $\displaystyle\int_0^1 \sqrt{1-x^2}\,dx = \frac{\pi}{4}$. **[4]**



The diagram shows the region R bounded by the ellipse with equation $4x^2 + (y-2)^2 = 4$ and the line $y = 2x$. The ellipse and the line intersect at the origin and the point $(1, 2)$.

(b) Find the area of R, giving your answer in terms of π. **[4]**

(c) Find the volume of solid generated when R is rotated 2π radians about the x-axis. **[2]**

Solution

Working follows the school's solution on grail.moe: https://grail.moe/library/18935/2025-yijc-prelim-h2-maths-papers-1-2-qp-soln

(a)

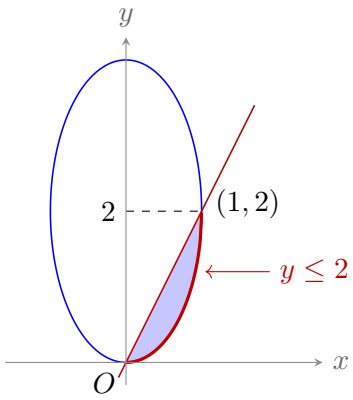
$$x = \sin\theta$$
$$\frac{dx}{d\theta} = \cos\theta$$

When $x = 0$, $\theta = 0$; $x = 1$, $\theta = \frac{\pi}{2}$

$$\begin{aligned}\int_0^1 \sqrt{1-x^2}\,dx &= \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta \\ &= \int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2}\cos 2\theta\,d\theta \\ &= \left[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{2}} \\ &= \left[\frac{1}{2}\left(\frac{\pi}{2}\right) + \frac{1}{4}\sin\pi\right] - \left[\frac{1}{2}(0) + \frac{1}{4}\sin 0\right] \\ &= \frac{\pi}{4}\end{aligned}$$

! Remember to change the limits when doing integration by substitution.
To get rid of the square root, use $1 - \sin^2\theta = \cos^2\theta$ (trigo identity).
Use MF27 (double angle formula) $\cos 2\theta = 2\cos^2\theta - 1$ to integrate $\cos^2\theta$.

(b)



$$\begin{aligned} 4x^2 + (y-2)^2 &= 4 \\ (y-2)^2 &= 4 - 4x^2 \\ y - 2 &= \pm\sqrt{4-4x^2} \\ y &= 2 \pm 2\sqrt{1-x^2} \end{aligned}$$

Since $y \leq 2$, $y = 2 - 2\sqrt{1-x^2}$

$$\begin{aligned} \text{Area of } R &= \frac{1}{2}(2)(1) - \int_0^1 y\,dx \\ &= 1 - \int_0^1 2 - 2\sqrt{1-x^2}\,dx \\ &= 1 - \int_0^1 2\,dx + 2\int_0^1 \sqrt{1-x^2}\,dx \\ &= 1 - \Big[2x\Big]_0^1 + 2\left(\frac{\pi}{4}\right) \\ &= \frac{\pi}{2} - 1 \end{aligned}$$

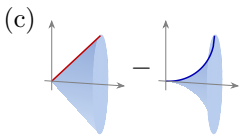
Alternatively

$$\text{Area of } R = \int_0^1 2x - \left(2 - 2\sqrt{1-x^2}\right)dx$$

(``upper curve'' − ``lower curve'')

! Observe from the graph that the y values on the curve that we are using to find the area under the curve are less than or equal to 2. Thus, we choose $y = 2 - 2\sqrt{1-x^2}$.

(c)



$$\begin{aligned} \text{Volume} &= \frac{1}{3}\pi(2)^2(1) - \pi\int_0^1 \left(2 - 2\sqrt{1-x^2}\right)^2 dx \\ &= 2.98 \text{ (3 s.f.)} \end{aligned}$$

Alternatively

$$\text{Volume} = \pi\int_0^1 (2x)^2\,dx - \pi\int_0^1 \left(2 - 2\sqrt{1-x^2}\right)^2 dx$$

! Required volume = Volume of cone − Volume generated by the region under the curve.
Volume of cone can be found by using $\frac{1}{3}\pi r^2 h$ or $\pi\int_0^1 (2x)^2\,dx$.

Question 7 10.5

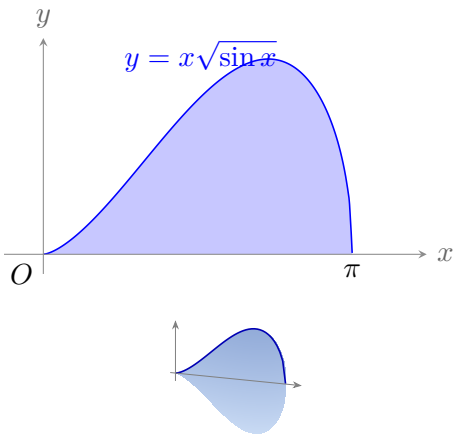
[NJC 2024 Promo Q2]

The region bounded by $y = x\sqrt{\sin x}$, the x-axis, and the lines $x = 0$ and $x = \pi$ is rotated through 2π radians about the x-axis. Find the exact volume of the solid generated. **[5]**

Solution

Working follows the school's solution on grail.moe: https://grail.moe/library/16719/njc-9758-2024-promo-solutions

$$\begin{aligned}
\int y^2\,dx &= \int x^2 \sin x\,dx \\
&= -x^2\cos x - \int 2x(-\cos x)\,dx \\
&= -x^2\cos x + 2\int x(\cos x)\,dx \\
&= -x^2\cos x + 2\left[x\sin x - \int \sin x\,dx\right] \\
&= -x^2\cos x + 2\left[x\sin x + \cos x\right] \\
&= -x^2\cos x + 2x\sin x + 2\cos x \\
&= \left(2 - x^2\right)\cos x + 2x\sin x + c
\end{aligned}$$



$$\begin{aligned}
\text{Volume required} &= \pi\int_0^{\pi} y^2\,dx \\
&= \pi\Big[\left(2 - x^2\right)\cos x + 2x\sin x\Big]_0^{\pi} \\
&= \pi\left[\left(2 - \pi^2\right)(-1) - (2)1\right] \\
&= \pi^3 - 4\pi
\end{aligned}$$

Question 8 10.6

[VJC 2025 Promo Q11(c), adapted]

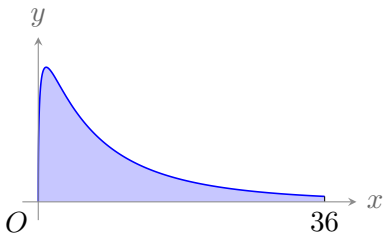
The parametric equations $x = t^2$, $y = 20te^{-t}$, for $0 \leq t \leq 6$, model a section of a roller coaster track; x is the horizontal distance and y the vertical distance from the ground, in metres. A wind-blocking panel is attached beneath the track. Its cross-sectional area is modelled by the area under the curve from $t = 0$ to $t = 6$. The area of the region under a parametric curve from $t = \alpha$ to $t = \beta$ is given by the formula $\displaystyle\int_{\alpha}^{\beta} y\left(\frac{dx}{dt}\right) dt$.

(i) Show that the cross-sectional area of the panel is $\displaystyle\int_{a}^{b} kt^2e^{-t}\,dt$, where a, b and k are integers to be determined. **[2]**

(ii) Find the exact area of the panel. **[4]**

Solution

(i)



$$\frac{dx}{dt} = 2t$$

$$\text{Area} = \int_0^6 y\left(\frac{dx}{dt}\right) dt$$

$$= \int_0^6 \left(20te^{-t}\right)(2t)\,dt$$

$$= \int_0^6 40t^2e^{-t}\,dt$$

$$\therefore\ k = 40, \quad a = 0, \quad b = 6 \text{ (shown)}$$

(ii)

$$\int t^2e^{-t}\,dt = -\int t^2(-1)e^{-t}\,dt$$

$$= -t^2e^{-t} - \int (2t)\left(-e^{-t}\right) dt$$

$$= -t^2e^{-t} + \int 2te^{-t}\,dt$$

$$= -t^2e^{-t} - 2te^{-t} - \int (2)\left(-e^{-t}\right) dt$$

$$= -t^2e^{-t} - 2te^{-t} + \int 2e^{-t}\,dt$$

$$= -t^2e^{-t} - 2te^{-t} - 2\int (-1)e^{-t}\,dt$$

$$= -t^2e^{-t} - 2te^{-t} - 2e^{-t} + C$$

$$= -e^{-t}\left(t^2 + 2t + 2\right) + C$$

$$\text{Area} = 40\int_0^6 t^2e^{-t}\,dt$$

$$= 40\Big[-e^{-t}\left(t^2 + 2t + 2\right)\Big]_0^6$$

$$= 40\left[-e^{-6}(36 + 12 + 2) - \left(-e^0(0 + 0 + 2)\right)\right]$$

$$= 40\left(-50e^{-6} + 2\right)$$

$$= 80 - 2000e^{-6} \text{ m}^2$$